

Avacort HFA 200mcg Metered Dose Inhaler

COMPOSITION:

Each metered dose inhaler contains 200 doses of Budesonide 200mcg per dose

PRODUCT DESCRIPTION:

Avacort HFA 200mcg Metered Dose Inhaler is a one-piece aluminum pressurized containers, containing liquid product under pressure, with no foreign or unpleasant odor, administered via a 50 µl metering valve. It contains 200 doses of 200 micrograms of budesonide per one metered dose. It contains ethanol, polyethylene glycol and Propellant HFA134a.

PHARMACOLOGICAL PROPERTIES:

Pharmacodynamic properties

ATC code : R03B A02

Budesonide is a glucocorticosteroid which possesses a high local anti-inflammatory action, with a lower incidence and severity of adverse effects than those seen with oral corticosteroids.

Pharmacokinetic properties

Absorption:

Following oral inhalation via Avacort HFA 200mcg Metered Dose Inhaler, peak plasma concentrations of budesonide (4.0 nmol/L after a dose of 800 µg) occur within 30 minutes. Maximum plasma concentration and area under the plasma concentration time profile increase linearly with dose, but are slightly (20-30%) higher following repeated doses (3 weeks treatment) than after a single dose. Lung deposition is estimated to 34% ±10% of the metered dose (arithmetic mean ± SD), while 22% will be retained in the mouthpiece and the rest (approximately 45% of the metered dose) will be swallowed.

The maximal plasma concentration after inhalation of 1 milligram budesonide is about 3.5 nmol/L and is reached after about 20 minutes.

Distribution

Budesonide has a volume of distribution of approximately 3 L/kg. Plasma protein binding averages 85-90%.

Metabolism:

Budesonide undergoes an extensive degree (approximately 90%) of biotransformation on first passage through the liver to metabolites of low glucocorticosteroid activity. The glucocorticosteroid activity of the major metabolites, 6β-hydroxybudesonide and 16α-hydroxyprednisolone, is less than 1% of that of budesonide. The metabolism of budesonide is primarily mediated by CYP3A, a subfamily of cytochrome p450.

Excretion:

The metabolites of budesonide are excreted as such or in conjugated form mainly via the kidneys. No unchanged budesonide has been detected in the urine. Budesonide has high systemic clearance (approximately 1.2 L/min), and the terminal half-life of budesonide after iv dosing averages 2-3 hours.

Paediatric safety data

Budesonide has a systemic clearance of approximately 0.5 L/min in 4-6 years old asthmatic children. Per kg body weight children have a clearance which is approximately 50% greater than in adults. The terminal half-life of budesonide after inhalation is approximately 2.3 hours in asthmatic children. This is about the same as in healthy adults.

INDICATIONS:

Avacort HFA 200mcg Metered Dose Inhaler is recommended in patients with bronchial asthma.

DOSAGE AND ADMINISTRATION:

Avacort HFA 200mcg Metered Dose Inhaler is for oral inhalation.

The dosage of Budesonide is individual and the maintenance dose should be the lowest possible.

Initially, at the beginning of inhaled corticosteroid therapy, for therapy during periods of severe asthma or when scaling down or withdrawing oral corticosteroids the dosage should be:

Children aged 2 - 7 years:

200-400 micrograms per day divided into 2-4 administrations.

Children aged 7 years and older:

200-800 micrograms per day divided into 2-4 administrations.

Adults:

400-1600 micrograms per day divided into 2-4 administrations.

For maintenance treatment administration twice daily, morning and evening, is usually sufficient.

After a single dose an effect may be expected after a few hours. The full therapeutic effect is only achieved after several weeks of treatment. Treatment with Budesonide is a prophylactic therapy with no demonstrated effect on acute disorders.

Patients dependent on oral steroids:

When transfer from oral steroids is initiated, the patient's condition must be relatively stable.

For 10 days a high dose of Budesonide is given in conjunction with the previously used dose of oral steroids. After that, the oral dose should be reduced gradually by e.g. 2.5 mg prednisolone or equivalent per month to the lowest possible level. The oral steroid can often be withdrawn completely.

Instructions for the correct use of Avacort HFA 200mcg Metered Dose Inhaler

Avacort HFA 200mcg Metered Dose Inhaler is inspiratory flow-driven which means that, when the patient inhales through the mouthpiece, the substance will follow the inspired air into the airways.

Note: It is important to instruct the patient:

- To carefully read the instructions for use in the patient information leaflet, which is packed with each inhaler
- To breathe in forcefully and deeply through the mouthpiece to ensure that an optimal dose is delivered to the lungs
- Never to breathe out through the mouthpiece
- To minimise the risk of oropharyngeal candida infection, the patient should rinse their mouth out with water after inhaling.

The patient may not taste or feel any medication when using the inhaler, due to the small amount of drug dispensed.

CONTRAINDICATIONS:

Hypersensitivity to budesonide or any components of Avacort HFA 200mcg Metered Dose Inhaler.

PRECAUTIONS AND WARNINGS

Precaution is necessary in patients with active or quiescent pulmonary tuberculosis, and in patients with fungal or viral infections in the airways.

Non steroid-dependent patients: A therapeutic effect is usually reached within 10 days. In patients with excessive mucus secretion in the bronchi, a short (about 2 weeks) additional oral corticosteroid regimen can be given initially.

Steroid-dependent patients: When transferred from oral steroids to Budesonide is started, the patient should be in a relatively stable phase. A high dose of Budesonide is then given in combination with the previously used oral steroid dose for about 10 days.

After that, the oral steroid dose should be gradually reduced (by for example 2.5 milligrams prednisolone or the equivalent each month) to the lowest possible level. In many cases, it is possible to completely substitute Budesonide for the oral steroid.

During transfer from oral therapy to Budesonide, a generally lower systemic steroid action will be experienced which may result in the appearance of allergic or arthritic symptoms such as rhinitis, eczema and muscle and joint pain. Specific treatment should be initiated for these conditions. During the withdrawal of oral steroids, patients may feel unwell in a non-specific way, even though respiratory function is maintained or improved. Patients should be encouraged to continue with Budesonide therapy whilst withdrawing the oral steroid, unless there are clinical signs to indicate the contrary. A general insufficient glucocorticosteroid effect should be suspected if, in rare cases, symptoms such as tiredness, headache, nausea and vomiting should occur. In these cases a temporary increase in the dose of oral glucocorticosteroids is sometimes necessary.

As with other inhalation therapy, paradoxical bronchospasm may occur, with an immediate increase in wheezing after dosing. If this occurs, treatment with inhaled budesonide should be discontinued immediately, the patient assessed and alternative therapy instituted if necessary.

Patients who have previously been dependent on oral steroids may, as a result of prolonged systemic steroid therapy, experience the effects of impaired adrenal function. Recovery may take a considerable amount of time after cessation of oral steroid therapy, hence oral steroid-dependent patients transferred to budesonide may remain at risk from impaired adrenal function for some considerable time. In such circumstances, HPA axis functions should be monitored regularly.

Acute exacerbations of asthma may need an increase in the dose of Budesonide or additional treatment with a short course of oral corticosteroid and/or an antibiotic, if there is an infection. The patient should be advised to use a short-acting inhaled bronchodilator as rescue medication to relieve acute asthma symptoms.

Budesonide inhalation is not intended for rapid relief of acute episodes of asthma where an inhaled short-acting bronchodilator is required.

If patients find short-acting bronchodilator treatment ineffective or they need more inhalations than usual, medical attention must be sought. In this situation consideration should be given to the need for or an increase in their regular therapy, e.g., higher doses of inhaled budesonide or the addition of a long-acting beta agonist, or for a course of oral glucocorticosteroid.

Patients, who have required high dose emergency corticosteroid therapy or prolonged treatment at the highest recommended dose of inhaled corticosteroids, may also be at risk of impaired adrenal function. These patients may exhibit signs and symptoms of adrenal insufficiency when exposed to severe stress. Additional systemic corticosteroid treatment should be considered during periods of stress or elective surgery. These patients should be instructed to carry a steroid warning card indicating their needs. Treatment with supplementary systemic steroids or Budesonide should not be stopped abruptly.

Systemic effects may occur with any inhaled corticosteroids, particularly at high doses prescribed for long periods. These effects are much less likely to occur with inhalation treatment than with oral corticosteroids. Possible systemic effects include Cushing's syndrome, Cushingoid features, adrenal suppression, growth retardation in children and adolescents, decrease in bone mineral density, cataract, glaucoma and more rarely, a range of psychological or behavioural effects including psychomotor hyperactivity, sleep disorders, anxiety, depression or aggression (particularly in children). It is important, therefore, that the dose of inhaled corticosteroid is titrated to the lowest dose at which effective control of asthma is maintained.

Influence on growth

It is recommended that the height of children receiving prolonged treatment with inhaled corticosteroids is regularly monitored. If growth is slowed, therapy should be re-evaluated with the aim of reducing the dose of inhaled corticosteroid, if possible, to the lowest dose at which effective control of asthma is maintained. The benefit of the corticosteroid therapy and the possible risk of growth suppression must be carefully weighed. In addition, consideration should be given to referring the patient to a paediatric respiratory specialist.

Reduced liver function affects the elimination of corticosteroids causing lower elimination rate and higher systemic exposure. Be aware of possible systemic side effects.

The plasma clearance following an intravenous dose of budesonide however was similar in cirrhotic patients and in healthy subjects. After oral ingestion systemic availability of budesonide was increased by compromised liver function due to decreased first pass metabolism. The clinical relevance of this to treatment with budesonide is unknown as no data exist for inhaled budesonide, but increases in plasma levels and hence an increased risk of systemic adverse effects could be expected.

Oral administration of ketoconazole and itraconazole (known inhibitors of CYP3A4 activity in the liver and in the intestinal mucosa may cause an increase in the systemic exposure to budesonide. Concomitant treatment with ketoconazole and itraconazole, HIV protease inhibitors or other potent CYP3A4 inhibitors should be avoided. If this is not possible, the period between treatments should be as long as possible. A reduction in the dose of budesonide should also be considered.

Oral candidiasis may occur during the therapy with inhaled corticosteroids. This infection may require treatment with appropriate antifungal therapy and in some patients discontinuation of treatment may be necessary.

Effects on ability to drive and use machines

Avacort HFA 200mcg Metered Dose Inhaler has no influence on the ability to drive and use machines.

PREGNANCY & LACTATION:

Pregnancy

Pregnancy category B. This is not likely to be relevant for humans given recommended doses, but therapy with inhaled Budesonide should be regularly reviewed and maintained at the lowest effective dose.

Lactation

It is not known whether budesonide passes into human breast milk. Administration of Avacort HFA 200mcg Metered Dose Inhaler to women who are breastfeeding should only be considered if the expected benefit to the mother is greater than any possible risk to the child.

SIDE EFFECT

Tabulated list of adverse reactions

The following definitions apply to the incidence of undesirable effects: Frequencies are defined as: very common ($\geq 1/10$); common ($\geq 1/100$ to $< 1/10$); uncommon ($\geq 1/1,000$ to $< 1/100$); rare ($\geq 1/10,000$ to $< 1/1,000$); very rare ($< 1/10,000$).

Table 1 Adverse Drug Reactions (ADR) by System Organ Class (SOC) and Frequency

SOC	Frequency	Adverse Drug Reaction
Infections and infestations	Common	Oropharyngeal candidiasis
Immune system disorders	Rare	Immediate and delayed hypersensitivity reactions including rash, contact dermatitis, urticaria, angioedema and anaphylactic reaction
Endocrine disorders	Rare	Signs and symptoms of

		systemic corticosteroid effects, including adrenal suppression and growth retardation*
Eye disorder	Unknown	Glaucoma Cataract
Psychiatric disorders	Rare	Psychomotor hyperactivity Sleep disorders Anxiety Depression Aggression Behavioural changes (predominantly in children)
Respiratory, thoracic and mediastinal disorders	Common	Cough Hoarseness Throat irritation
	Rare	Bronchospasm Dysphonia Hoarseness**
Skin and subcutaneous tissue disorders	Rare	Bruising

* refer to Paediatric population below

** rare in children

Description of selected adverse reactions

The candida infection in the oropharynx is due to drug deposition. Advising the patient to rinse the mouth out with water after each dosing will minimise the risk.

As with other inhalation therapy, paradoxical bronchospasm may occur in very rare cases (see paragraph 'WARNINGS AND PRECAUTIONS').

Paediatric population

Due to the risk of growth retardation in the paediatric population, growth should be monitored as described under paragraph 'WARNINGS AND PRECAUTIONS'.

DRUG INTERACTIONS:

The metabolism of budesonide is primarily mediated by CYP3A4. Inhibitors of this enzyme, e.g. ketoconazole and itraconazole, can therefore increase systemic exposure to budesonide. The combination should be avoided. If this is not possible, the period between treatments should be as long as possible and a reduction of the budesonide dose could also be considered. Other potent inhibitors of CYP3A4 are also likely to markedly increase plasma levels of budesonide.

Marked increases in plasma levels (on average four- fold) may occur if itraconazole, 200 mg once daily, is administered concomitantly with inhaled budesonide (single dose of 1000 µg).

Raised plasma concentrations of and enhanced effects of corticosteroids have been observed in women also treated with oestrogen and contraceptive steroids, but no effect has been observed with budesonide and concomitant intake of low dose combination oral contraceptives.

Because adrenal function may be suppressed, an ACTH stimulation test for diagnosing pituitary insufficiency might show false results (low values).

OVERDOSAGE:

Acute overdosage with Avacort HFA 200mcg Metered Dose Inhaler, even in excessive doses, is not expected to be a clinical problem. The only harmful effect that follows inhalation of large amounts of the drug over a short period is suppression of hypothalamic-pituitary-adrenal (HPA) function.

No special emergency action needs to be taken. Treatment with Avacort HFA 200mcg Metered Dose Inhaler should be continued at the recommended dose to control the asthma.

STORAGE:

Store below 30°C. Protect from light and moisture.
Keep out of reach of children.

PACKING:

One unit metered dose inhaler is packed in a printed box.

Product Registration holder:

Nano Medic Care Sdn Bhd
No. 1-6 & 1-12, 1st floor, Jalan Indah 2, Taman Indah,
84000 Muar, Johor, Malaysia

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Synmosa Biopharma Corporation, Synmosa Plant,
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